

Customer Segmentation & Targeted Campaigns

Machine Learning II — NOVA IMS — project notes — 08 Jun 2026

The gist

We have a supermarket's customer database and a simple goal: stop treating everyone the same. The project splits ~33,000 customers into a handful of look-alike groups, gives each group a plain-English name and profile, and then suggests a marketing campaign tailored to what that group actually buys. It ships as an interactive Streamlit app (the course lets you hand in a working app instead of a written report), so a marketer can poke at it live.

The data

Two CSVs. **customer_info** has one row per customer — lifetime spend across 10 product categories (groceries, meat, electronics, videogames, pet food, ...), how promotion-driven they are, how many distinct products/stores they touch, plus demographics (age from birthdate, gender, education parsed from the name, kids at home, loyalty card). **customer_basket** is the transaction log — the list of goods in each basket — used later for product-association rules. A few values are missing here and there; those get median-imputed so nothing crashes.

A quick look at the data

Before clustering, a sanity check on what's in there. Groceries dominate total spend by a wide margin, and the category spends are heavily right-skewed (a long tail of big spenders) — which is exactly why they get log-scaled before clustering. Age is roughly bell-shaped and centred in the low 50s.

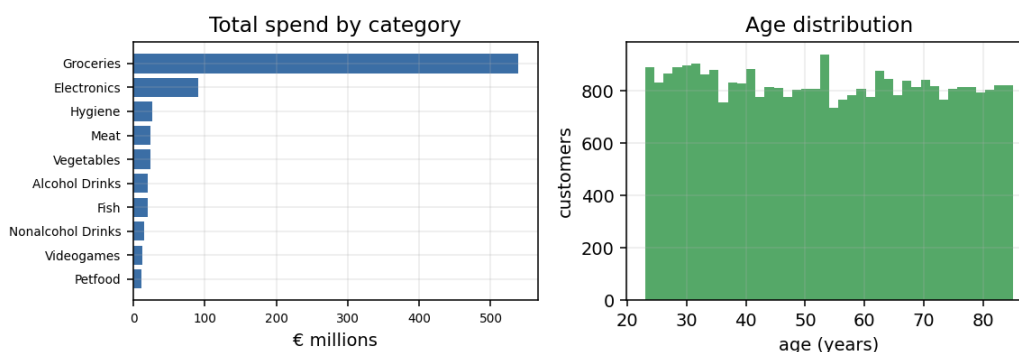


Figure 1 — total spend per category (left) and the age distribution (right).

How the segmentation works

The whole approach hinges on one choice: **cluster on behaviour, profile on demographics**. K-Means is given only how people *shop* — the 10 category spends (log-scaled), how promotion-driven they are, and how wide their product/store footprint is. Who they are (age, gender, education, family) is deliberately kept out of the distance and layered back on afterwards to describe each segment. We sweep K from 2 to 8; the silhouette score peaks at **K = 4** and the inertia elbow agrees, so that's the default.

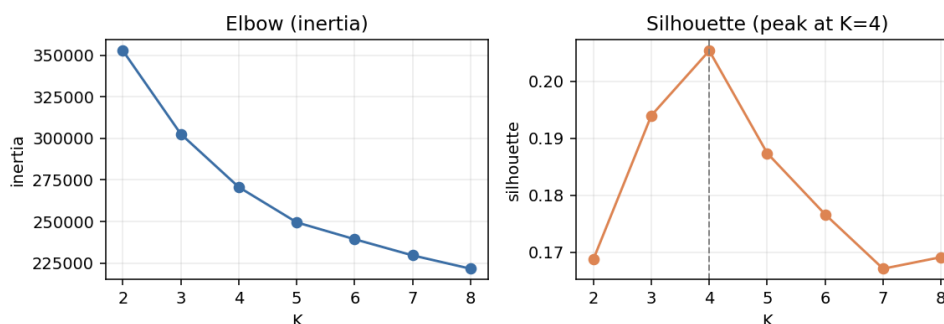


Figure 2 — picking K. Silhouette (right) is the deciding metric; dashed line marks the peak.

The four segments

Names are **generated from each segment's own profile** (its over-indexed categories plus tags like Promo-Driven / Large-Family / Power), never hard-coded — so a label can't contradict the data. Biggest spenders first:

Segment	Size	Avg spend	% on promo
Large-Family Power Grocery Shoppers	29%	€36,489	32%
Gaming & Electronics Shoppers	12%	€22,760	26%
Promo-Driven Grocery Shoppers	37%	€17,506	45%
Veggie & Hygiene Shoppers	22%	€17,400	13%

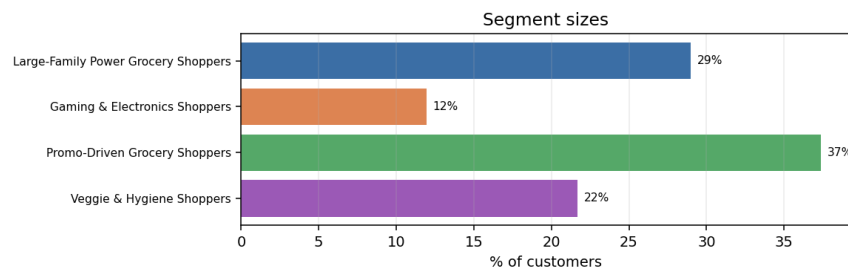


Figure 3 — how the customer base splits.

Key decisions (and why)

The reasoning behind the calls that actually shaped the result — these are the bits worth defending if anyone asks:

Behaviour in, demographics out. Mixing age/gender/education/kids into the distance let loosely-related, noisy attributes drag the clusters around — and one cluster ended up swallowing ~46% of customers. Pulling demographics out (and using them only to profile) balanced the sizes and even moved the silhouette-best K from 3 to 4.

Drop *total_spend* from the inputs. It is literally the sum of the ten category columns, so feeding it alongside them double-counts overall magnitude and lets pure “big vs small spender” dominate everything. It is still computed — just used for profiling, not for the distance.

Log-scale the spends. Category spend is heavily right-skewed; without a log transform a few huge spenders own the Euclidean distance and the clustering reacts to *amount* rather than *pattern*. A log1p fixes that, then everything is standardised so no single feature shouts over the others.

Median-impute, don't drop rows. Missing values are filled with column medians instead of dropping customers, so the app survives a stray blank from a clean environment and every customer still ends up in the exported cluster file (a grading requirement).

Data-driven names, not hard-coded labels. The first draft pinned names to cluster ids — and the cluster it called “Budget” actually had the *highest* average spend. Deriving names from each segment's signature means the label always matches the data and survives re-runs or a different K.

One source of truth for names and campaigns. Both come from the same segment signature, so the offer can never disagree with the description — a family-heavy grocery segment gets a family bulk deal, not some unrelated promo.

Guard against duplicate names. At higher K two clusters can derive the same name; a numeric suffix keeps them distinct so the profile heatmap and radar don't collide on a duplicate index.

Segment profiles & campaigns

Each segment in plain English, followed by the offers the app actually generates for it. Because the offers are built from the same profile as the description, they always match the segment they target:

Large-Family Power Grocery Shoppers

29% of customers; ~€36,489 avg lifetime spend; over-indexes on grocery; large families (~3.2 kids/teens at home); the highest-value segment.

- **Family Bulk Deal** — 15% off when the basket exceeds €120
- **Grocery Loyalty Boost** — double loyalty points on all grocery purchases this month

Gaming & Electronics Shoppers

12% of customers; ~€22,760 avg lifetime spend; over-indexes on gaming & electronics.

- **Gaming + Electronics Bundle** — buy both together, get 20% off the cheaper one
- **Gaming Loyalty Boost** — double loyalty points on all gaming purchases this month

Promo-Driven Grocery Shoppers

37% of customers; ~€17,506 avg lifetime spend; over-indexes on grocery; highly promotion-driven (45% of buys on promo).

- **Grocery Loyalty Boost** — double loyalty points on all grocery purchases this month
- **Weekly Promo Drop** — exclusive extra discount on top categories every week (this segment buys heavily on promotion)

Veggie & Hygiene Shoppers

22% of customers; ~€17,400 avg lifetime spend; over-indexes on veggie & hygiene.

- **Veggie + Hygiene Bundle** — buy both together, get 20% off the cheaper one
- **Veggie Loyalty Boost** — double loyalty points on all veggie purchases this month

On the silhouette score

The clusters score a silhouette of ~0.21 at $K = 4$. On the $-1 \dots 1$ scale that's *weak-but-real* structure — exactly what retail spend gives you: it's continuous and overlapping, with no natural hard borders between shopper types. We did check whether the number could be pushed up by clustering in a PCA-compressed space. It can — but only by throwing information away:

Clustering space	Info kept	Silhouette
Full behavioural (used)	100%	0.205
PCA, 8-D	82%	0.256
PCA, 5-D	68%	0.322
PCA, 3-D	53%	0.423
PCA, 2-D	40%	0.423

So the prettier numbers are a measurement artefact — silhouette mechanically rises as you drop dimensions, because distances get tamer in low-D. You're not finding better segments, just measuring in a kinder space; and clustering on 2 components (40% of the signal) would blur the per-category spend the names and campaigns lean on. We kept the full behavioural space on purpose: honest, interpretable segments beat a flattering score.

A note on the environment

The app runs in its own isolated environment with every library pinned to a tested version. For clustering this matters more than it sounds — K-Means results depend on the exact NumPy / scikit-learn build, so an unpinned setup can hand you a *different* set of segments from the same code. Pinning plus isolation is what lets a grader reproduce these exact four segments. The environment itself isn't shipped — it's a local sandbox, rebuilt in seconds from the pinned list.

Wrap-up

Four behavioural segments fall out of the data cleanly (silhouette peaks at $K = 4$), each with a name, a profile and a matching campaign straight from how that group shops — nothing hard-coded. The takeaway: *behaviour*, not demographics, carves this base into useful groups — the biggest spenders aren't a particular age or degree, they're the large families filling big grocery baskets, and that's where the budget should go.